📘 AI/ML Internship – Day 35 Notes & Tasks

🟢 Module 7: Introduction to Scikit-learn & First ML Model

📌 Part 1: What is Scikit-learn?

🔹 Definition

👉 Scikit-learn is a Python library used for:

- Machine Learning models
- Data preprocessing
- Training & prediction

👉 It is one of the most popular ML libraries.

📌 Part 2: Why Scikit-learn is Important?

👉 Scikit-learn provides:

- Ready-made ML algorithms
- Easy model training
- Simple prediction methods

👉 Used in:

- AI projects
- Data Science
- Prediction systems

📌 Part 3: Theory Answers (IMPORTANT)

🔹 1. What is Scikit-learn?

👉 Scikit-learn is a Python library used for building Machine Learning models.

🔹 2. Why Scikit-learn is Popular?

👉 Because:

- Easy syntax
- Many ML algorithms
- Fast model development

🔹 3. What is a Machine Learning Model?

👉 ML model is a system trained using data to make predictions.

🔹 4. What is Training Data?

👉 Data used to teach the model.

🔹 5. What is Prediction?

👉 Output generated by trained model.

📌 Part 4: Installing Scikit-learn

```
pip install scikit-learn
```

📌 Part 5: Import Libraries

```
import pandas as pd
from sklearn.linear_model import LinearRegression
```

📌 Part 6: Understanding First ML Problem

🔹 Problem Statement

👉 Predict student marks based on study hours.

🔹 Dataset

Study Hours	Marks
1	20
2	40
3	60
4	80
5	100

📌 Part 7: Features & Labels

Term	Meaning
Feature	Input data
Label	Output data

🔹 Example

Feature → Study Hours
Label → Marks

📌 Part 8: Create Dataset

```
data = {
    "Hours": [1, 2, 3, 4, 5],
    "Marks": [20, 40, 60, 80, 100]
}

df = pd.DataFrame(data)

print(df)
```

📌 Part 9: Prepare Input & Output

X = df[["Hours"]]

y = df["Marks"]

👉 X = input/features
👉 y = output/labels

📌 Part 10: Create Linear Regression Model

model = LinearRegression()

📌 Part 11: Train the Model

model.fit(X, y)

👉 fit() means training the model.

📌 Part 12: Make Prediction

prediction = model.predict([[6]])

print(prediction)

👉 Predict marks for 6 study hours.

📌 Part 13: Complete ML Program

```
import pandas as pd
from sklearn.linear_model import LinearRegression

data = {
   "Hours": [1, 2, 3, 4, 5],
   "Marks": [20, 40, 60, 80, 100]
}
```

```
df = pd.DataFrame(data)

X = df[["Hours"]]
y = df["Marks"]

model = LinearRegression()

model.fit(X, y)

prediction = model.predict([[6]])

print("Predicted Marks:", prediction)
```

📌 Part 14: Visualization of ML Flow

Dataset
↓
Train Model
↓
Learn Pattern
↓
Prediction

📌 Part 15: Real-World Thinking

👉 Linear Regression is used for:

- House price prediction
- Salary prediction
- Sales forecasting
- Weather analysis

📌 Part 16: Why This is Important?

👉 Today students build:
✅ First ML model
✅ First prediction system

👉 This is the starting point of real AI development.

🎯 Tasks for Day 35

🧠 Task 1: Theory (Write Answers)

1. What is Scikit-learn?
2. What is ML model?
3. What is feature?
4. What is label?
5. What is prediction?

💻 Task 2: Dataset Practice

1. Create student dataset
2. Create salary dataset
3. Create sales dataset

💻 Task 3: Model Training

1. Train linear regression model
2. Use fit()
3. Understand learning process

💻 Task 4: Prediction Practice

1. Predict marks
2. Predict salary

3. Predict sales values

💻 Task 5: Feature & Label Identification

1. Identify input/output
2. Explain X and y
3. Explain training logic

💻 Task 6: Real-World ML Examples

1. House price prediction
2. Weather prediction
3. Employee salary prediction